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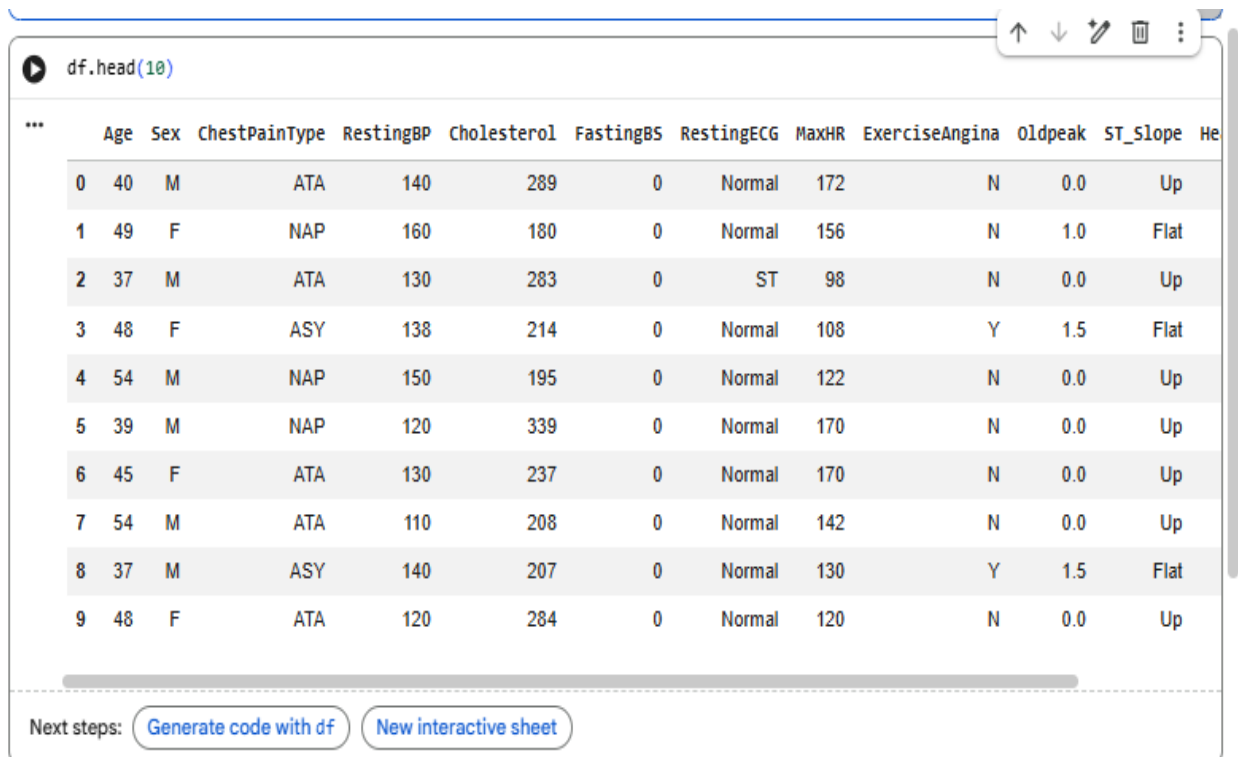
Clg Name :- T.B.Girwalkr Polytechnic , Ambajogai

Domain :- AI - ML

Branch Name :- Computer Engineering

- 1) Load the Heart Disease dataset using pandas and display the first 10 rows. Also check the shape and basic information (info()) of the dataset.

A) 10 rows



The screenshot shows a Jupyter Notebook interface with a code cell containing `df.head(10)`. The output is a table with 13 columns: Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope, and HeartDisease. The first 10 rows of data are displayed.

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	1
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	1
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	1
5	39	M	NAP	120	339	0	Normal	170	N	0.0	Up	1
6	45	F	ATA	130	237	0	Normal	170	N	0.0	Up	1
7	54	M	ATA	110	208	0	Normal	142	N	0.0	Up	1
8	37	M	ASY	140	207	0	Normal	130	Y	1.5	Flat	1
9	48	F	ATA	120	284	0	Normal	120	N	0.0	Up	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

B) Shape

```
df.shape
```

```
*** (918, 12)
```

C) Info

```
df.info()
```

```
*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   918 non-null   int64
1   Sex                   918 non-null   object
2   ChestPainType         918 non-null   object
3   RestingBP             918 non-null   int64
4   Cholesterol            918 non-null   int64
5   FastingBS             918 non-null   int64
6   RestingECG            918 non-null   object
7   MaxHR                 918 non-null   int64
8   ExerciseAngina        918 non-null   object
9   Oldpeak               918 non-null   float64
10  ST_Slope              918 non-null   object
11  HeartDisease          918 non-null   int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB
```

2) Check for missing values in the dataset. Show the count of null values for each column.

```
df.isnull().sum()
```

Age	0
Sex	0
ChestPainType	0
RestingBP	0
Cholesterol	0
FastingBS	0
RestingECG	0
MaxHR	0
ExerciseAngina	0
Oldpeak	0
ST_Slope	0
HeartDisease	0

dtype: int64

- 3) Check for duplicate rows in the dataset. If any duplicates are found, remove them and print the new shape of the dataset.

```
df.duplicated().sum()
... np.int64(0)
```

- 4) Create a function to plot histograms for the following numerical columns:

Age

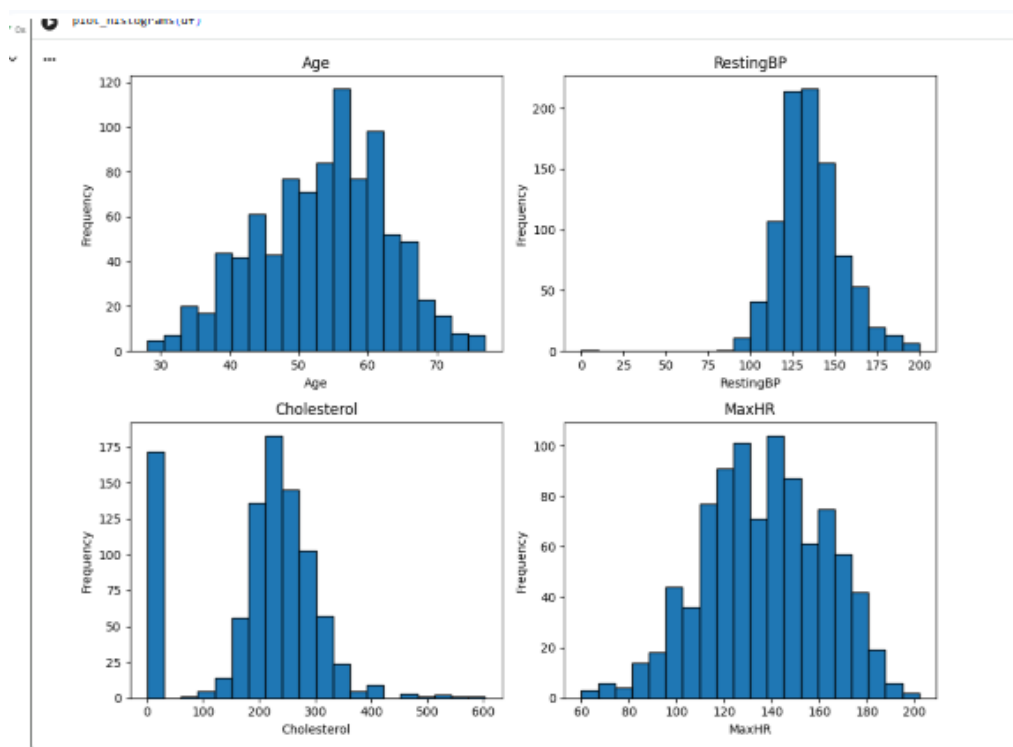
RestingBP

Cholesterol

MaxHR

Plot all four histograms in one figure using subplots (2x2 layout).

Use this function to visualize the data after cleaning.



- 5) Perform One-Hot Encoding on all categorical columns using `pd.get_dummies()`.
Store the result in a new dataframe called `df_encoded`.
Print the shape and the first 5 rows of the encoded dataframe.

```
*** Shape: (918, 21)
```

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisease	Sex_F	Sex_M	ChestPainType_ASY	...	ChestPainType_NAP
0	40	140	289	0	172	0.0	0	False	True	False	...	False
1	49	160	180	0	156	1.0	1	True	False	False	...	True
2	37	130	283	0	98	0.0	0	False	True	False	...	False
3	48	138	214	0	108	1.5	1	True	False	True	...	False
4	54	150	195	0	122	0.0	0	False	True	False	...	True

5 rows x 21 columns

- 6) After cleaning and encoding:
Print the final shape of `df_encoded`.
Show the list of all column names in the final dataframe.

```
*** Final Shape: (918, 21)
Column Names:
Index(['Age', 'RestingBP', 'Cholesterol', 'FastingBS', 'MaxHR', 'Oldpeak',
      'HeartDisease', 'Sex_F', 'Sex_M', 'ChestPainType_ASY',
      'ChestPainType_ATA', 'ChestPainType_NAP', 'ChestPainType_TA',
      'RestingECG_LVH', 'RestingECG_Normal', 'RestingECG_ST',
      'ExerciseAngina_N', 'ExerciseAngina_Y', 'ST_Slope_Down',
      'ST_Slope_Flat', 'ST_Slope_Up'],
      dtype='object')
```

7) Ans :-

- a) Cholesterol = 0 and RestingBP=0
- b) Replaced zero value with the mean of non – zero value
- c) Invalid value can affect analysis and reduce model accuracy
- d) Convert categorical data into numerical form for machine learning
- e) Dataset became cleaner and ready for model training